

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING (5 Questions)

Course Code:	ITA05	Course Title:	COMPUTER VISION
CO Assessed:	CO2 – Manipulation(BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	20% of CIA	Total Marks:	50 (5 Questions × 10 Marks)
CO2 Statement:	Apply image enhancement and segmentation techniques for extracting meaningful regions from images..(BL3)		
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Section / Batch:	Slot B / 2024	Date:	28/07/26

Code of Conduct
 I, Nagendra (192472399) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Nagendra Reddy

- Instructions**
- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
 - No negative marking. Unanswered questions carry zero marks.
 - No calculators or electronic devices permitted.
 - Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Grey Level Transformations	Enhancing image brightness and contrast using negative, log, power-law (gamma), and contrast stretching transformations.	1	10
Histogram Processing Techniques	Improving image contrast through histogram analysis, histogram equalization, and histogram matching.	2	10
Spatial Domain Enhancement	Reducing noise and enhancing image quality using smoothing filters and sharpening filters.	3	10
Frequency Domain Enhancement	Enhancing images by applying low-pass and high-pass filters.	4	10
Image Segmentation	Separating objects from the background using point, line, and edge detection, thresholding, region-based segmentation, edge linking, and boundary detection techniques.	5	10
GRAND TOTAL:			50 Marks

Annexure A – ANALYTICAL PROBLEM SOLVING

1. Grey Level Transformation

A grayscale image is enhanced using a negative transformation.

(a) Explain the purpose of grey level transformation in image enhancement

(b) Apply the negative transformation:

$$s = 255 - r$$

to the following pixel values:

$$r = 50, 100, 150, 200$$

Compute the transformed pixel values and interpret the effect on the image.

2. Histogram Equalization

A 4-bit grayscale image has the following histogram:

Intensity	Frequency
0	4
1	6
2	10
3	6

(a) Explain why histogram equalization is used for image enhancement.

(b) Compute the cumulative distribution function (CDF).

(c) Determine the equalized intensity values and interpret the improvement in image contrast.

3. Gaussian Smoothing

A Gaussian filter is applied to reduce image noise.

(a) Explain why Gaussian filtering is preferred over averaging filtering.

(b) Apply the Gaussian kernel

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 & 2 & 4 & 2 & 1 & 2 & 1 \end{bmatrix}$$

to the image region

$$\begin{bmatrix} 10 & 20 & 10 & 20 & 40 & 20 & 10 & 20 & 10 \end{bmatrix}$$

Compute the new center pixel value and interpret the smoothing effect.

4. Edge Detection Using Sobel Operator

An image is processed using the horizontal Sobel operator.

(a) Explain the concept and purpose of edge detection.

(b) Apply the horizontal Sobel mask

$$[-1 \ -2 \ -1 \ 0 \ 0 \ 1 \ 2 \ 1]$$

to the image

$$[10 \ 10 \ 10 \ 20 \ 20 \ 20 \ 30 \ 30 \ 30]$$

Compute the gradient at the center and interpret the result.

5. Contrast Stretching

A grayscale image has intensity values in the range 50–150.

(a) Explain the need for contrast stretching in image enhancement.

(b) Using linear contrast stretching to map the range [50,150] to [0,255], compute the transformed values for

$$r = 50, 100, 150$$

Show the calculations and interpret the improvement in image contrast.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks	
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem		

Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method		
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process		
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations		
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

a) Purpose of Gray level transformation:

Gray level transformation improves the visual quality of an image by modifying pixel intensity values. It enhances brightness, contrast, and highlights important image details.

b) Negative transformation:

Formula: $s = 255 - r$

Transformed pixel ($s = 255 - \text{original pixel } (r)$)

original pixel (r)	Calculation	New Pixel (s)
		205
50	$255 - 50$	155
100	$255 - 100$	105
150	$255 - 150$	55
200	$255 - 200$	

Transformed pixels are: 205, 155, 105, 55.

Interpretation:

The negative transformation reverses brightness value. Bright pixel becomes dark and dark pixel becomes bright, making hidden details more visible.

a) Histogram Equalization: Histogram equalization improves image contrast by spreading pixel intensities over the available intensity range.

b) Cumulative distribution Function (CDF):

CDF =

$$\frac{\text{No. of Pixels}}{\text{Total Pixels}}$$

Total Pixels (N) = $4 + 6 + 10 + 6 = 26$

Intensity	Frequency	CDF
0	4	4
1	6	10
2	10	20
3	6	26

a) Equalized intensity values :

Formula : $s = \frac{CDF}{N} \times (L-1)$

Where $L = 4$, so $L-1 = 3$.

Intensity	CDF	Equalized value
0	4	$\frac{4}{26} \times 3 = 0.46 \approx 0$
1	10	$\frac{10}{26} \times 3 = 1.15 \approx 1$
2	20	$\frac{20}{26} \times 3 = 2.31 \approx 2$
3	26	$\frac{26}{26} \times 3 = 3$

Interpretation :-

Histogram equalization spreads the intensity values more uniformly increasing image contrast and making image details clear

1) Gaussian Filtering :

Removes noise more effectively.

Produces Smoother images with less blurring.

Preserves edges better than an averaging filter.

2) Calculation :

Gaussian kernel :

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Image :

$$\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

Multiply Corresponding values :

$$1(10) + 2(20) + 1(10) + 2(20) + 4(40) + 2(20) + 1(10) + 2(20) + 1(10)$$

$$= 10 + 40 + 10 + 40 + 160 + 40 + 10 + 40 + 10 = 360$$

$$\text{New Center Pixel} = \frac{360}{16} = 22.5 \approx 23 //$$

Interpretation :

The center Pixel changes from 40 to 23, reducing noise and smoothing the image while preserving its overall appearance.

④ a) Purpose:

Edge detection identifies object boundaries by detecting sudden intensity changes in an image.

b) Calculation:

Horizontal Sobel mask:

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

image:

$$\begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

$$= (-1 \times 10) + (-2 \times 10) + (-1 \times 10) + (0) + (1 \times 30) + (2 \times 30) + (1 \times 30)$$

$$= -10 - 20 - 10 + 30 + 60 + 30 = 80$$

Gradient = 80

Interpretation:

A gradient value of 80 indicates a strong horizontal edge, showing a significant intensity change from top to bottom.